

1.5 Lines in 3-Space Worksheet

1. Determine whether each of the following statements are true or false.

- a) Two lines that are parallel to a third line are parallel to each other.

- b) Two lines that are perpendicular to a third line are parallel to each other.

- c) Two lines either intersect or are parallel.

- 2. Determine the vector equation and parametric equation for the line.
- a) The line through the point $(6, -5, 2)$ and parallel to the vector $\left\langle 1, 3, -\frac{2}{3} \right\rangle$.

b) The line through the point $(2, 2.4, 3.5)$ and parallel to the vector $3\mathbf{i} + 2\mathbf{j} - \mathbf{k}$.

c) The line through the point $(0, 14, -10)$ and parallel to the line $\begin{cases} x = -1 + 2t \\ y = 6 - 3t \\ z = 3 + 9t \end{cases}$.

d) The line through the point $(1, 0, 6)$ and perpendicular to the line $\begin{cases} x = -1 + 2t \\ y = 6 - 3t \\ z = 3 + 9t \end{cases}$.

e) The line passing through the points $(2, 0, 3)$ and $(3, 1, 6)$.

3. Which of the following four lines are parallel? Are any of them identical?

$$\begin{aligned} \mathbf{L}_1 &= \langle 1+6t, 1-3t, 12t+5 \rangle \\ \mathbf{L}_2 &= \langle 1+2t, t, 1+4t \rangle \end{aligned} \quad L_3 = \begin{cases} x = 3+4t \\ y = 1+2t \\ z = 5+8t \end{cases} \quad L_4 = \begin{cases} t = 2x-2 \\ t = 4-4y \\ t = z+1 \end{cases}$$

4. Determine whether the given lines are parallel, skew, or intersect. If they intersect, determine their point of intersection.

$$L_1(t) = \begin{cases} x = 3t \\ y = t \\ z = -1+t \end{cases} \quad L_2(t) = \begin{cases} x = t \\ y = 2-t \\ z = -2+t \end{cases}$$

5. Determine whether the given lines are parallel, skew, or intersect. If they intersect, determine their point of intersection.

$$L_1(t) = \begin{cases} x = 3t \\ y = t \\ z = -1 + t \end{cases} \quad L_2(t) = \begin{cases} x = 1 + t \\ y = -1 + t \\ z = 5 + t \end{cases}$$

6. Consider the vectors $\mathbf{a} = \langle 0, 1, 1 \rangle$ and $\mathbf{b} = \langle -1, 0, 1 \rangle$.

a) Find $\text{proj}_{\mathbf{a}}(\mathbf{b})$, the vector projection of $\mathbf{b}$ on $\mathbf{a}$.

b) Find the distance from the point $(-1, 0, 1)$ to the line through the origin whose direction vector is $\mathbf{a}$.

7. Determine each of the following distances between the point and line.

a) The distance between the point $(4, 1, -2)$ and the line $L = \begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = 4 - 3t \end{cases}$.

b) The distance between the point $(0, 1, 3)$ and the line $\mathbf{L}(t) = \langle 2t, 6 - 2t, 3 + t \rangle$.

c) The distance between the two parallel lines $L_1 = \begin{cases} x = 1 + t \\ y = 3 - 2t \\ z = 4 - 3t \end{cases}$ and $L_2 = \begin{cases} x = 2 + 2t \\ y = 7 - 4t \\ z = 5 - 6t \end{cases}$.